

CSCI 1101 Introduction to Computer Science

Digital Logic

Overview

Introduction

- Logic Circuits
- Logic Gates
- Boolean Algebra
- Truth Tables

Types of Logic Gates

- NOT
- AND and NAND
- OR and NOR
- XOR and XNOR

Circuits

- Circuits
- Combinational Circuits
- Example 1

Boolean Algebra Rules

- Axioms of Boolean Algebra
- Single-Variable Rules (Theorems)
- Two- and Three- Variable Properties
- Order of Operations

Circuit and Boolean Expression Equivalence and Simplification

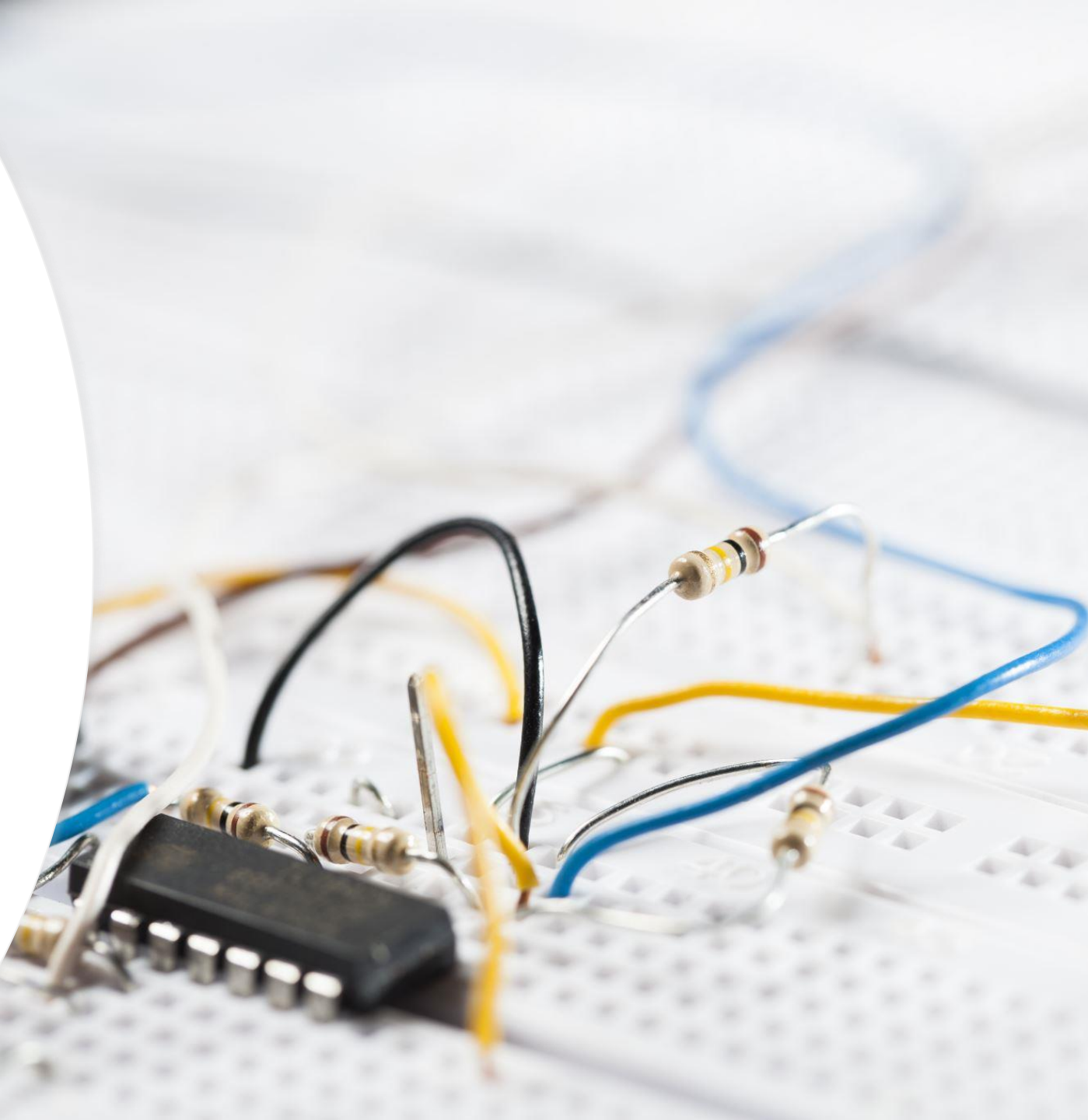
- Circuit Equivalence
- XOR Gate Equivalence
- Example 2

Applications of Circuits

- Constructing Gates
- Integrated Circuits
- CPU Chips

Computers and Circuits

- Computers are constructed from logic circuits
 - Logic circuits perform operations of digital signals
 - Usually implemented as circuits where signal values are restricted to a few values (e.g., binary circuits)
- Logic circuits are logic gates combined to perform more complicated tasks





What are logic gates?

- Logic gates are the switches that turn ON or OFF depending on what the user is doing (status of condition).
 - They check whether or not the information they get follows a certain rule.
- They are the building blocks for how computers work.

True = ON = 1

False = OFF = 0

Boolean Algebra

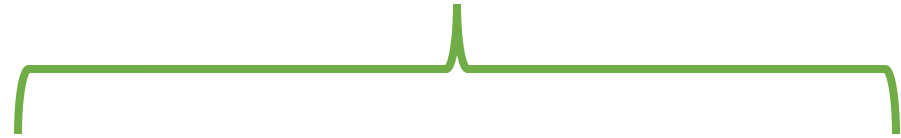
- A scheme for describing in algebra the processes involved in logical reasoning
- Provides a means of describing circuits built with switches (logic gates)
- Works with variables that can have only 2 values – true or false (1 or 0, respectively)
- A tool to simplify logical circuits

Truth Table

A truth table is a means for describing how a logic circuit's output depends on the logic levels present at the circuit's inputs.

Number of Rows = 2^{inputs}

Inputs



A	B	C		OUTPUT
0	0	0	...	0/1
0	0	1	...	0/1
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.

Name	NOT	AND	NAND	OR	NOR	XOR	XNOR																																																																																																
Alg. Expr.	$\overline{A}$	AB	$\overline{AB}$	$A + B$	$\overline{A + B}$	$A \oplus B$	$\overline{A \oplus B}$																																																																																																
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Truth Table	<table><tr><th>A</th><th>X</th></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td></tr></table>	A	X	0	1	1	0	<table><tr><th>B</th><th>A</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	B	A	X	0	0	0	0	1	0	1	0	0	1	1	1	<table><tr><th>B</th><th>A</th><th>X</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	B	A	X	0	0	1	0	1	1	1	0	1	1	1	0	<table><tr><th>B</th><th>A</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	B	A	X	0	0	0	0	1	1	1	0	1	1	1	1	<table><tr><th>B</th><th>A</th><th>X</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	B	A	X	0	0	1	0	1	0	1	0	0	1	1	0	<table><tr><th>B</th><th>A</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	B	A	X	0	0	0	0	1	1	1	0	1	1	1	0	<table><tr><th>B</th><th>A</th><th>X</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	B	A	X	0	0	1	0	1	0	1	0	0	1	1	1
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Types of Logic Gates

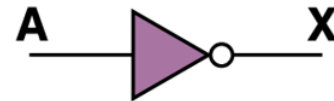
NOT Gate

A NOT gate accepts one input signal (0 or 1) and returns the opposite signal as output.

Boolean Expression:

- $X = A'$
- $X = \bar{A}$
- $X = \neg A$

Logic Diagram Symbol



Truth Table

A	X
0	1
1	0

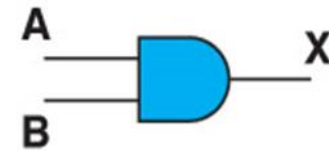
AND Gate

An AND gate accepts two inputs signals. If both are 1, the output is 1; otherwise, the output is 0.

Boolean Expression:

- $X = A \wedge B$
- $X = A \cdot B$
- $X = AB$

Logic Diagram Symbol



Truth Table

A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

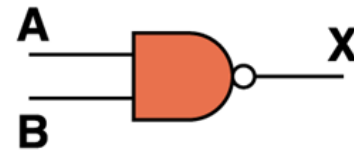
NAND Gate

The NAND gate accepts two input signals. If both are 1, the output is 0; otherwise, the output is 1.

Boolean Expression:

- $X = \overline{A \wedge B} = (A \wedge B)' = \neg(A \wedge B)$
- $X = \overline{A \cdot B} = (A \cdot B)' = \neg(A \cdot B)$
- $X = \overline{AB} = (AB)' = \neg(AB)$

Logic Diagram Symbol



Truth Table

A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

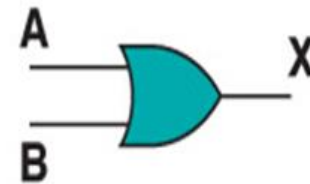
OR Gate

An OR gate accepts two input signals. If both are 0, the output is 0; otherwise, the output is 1.

Boolean Expression:

- $X = A + B$
- $X = A \vee B$

Logic Diagram Symbol



Truth Table

A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

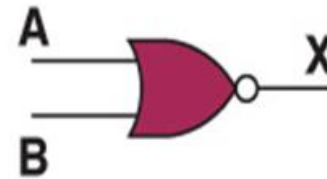
NOR Gate

An NOR gate accepts two input signals. If both are 0, the output is 1; otherwise, the output is 0.

Boolean Expression:

- $X = \overline{A + B} = (A + B)' = \neg(A + B)$
- $X = \overline{A \vee B} = (A \vee B)' = \neg(A \vee B)$

Logic Diagram Symbol



Truth Table

A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

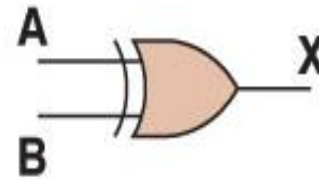
XOR Gate

A XOR (exclusive OR) gate accepts two input signals. If both are the same, the output is 0; otherwise, the output is 1.

Boolean Expression:

- $X = A \oplus B$

Logic Diagram Symbol



Truth Table

A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

XNOR Gate

A XNOR gate accepts two input signals. If both are the same, the output is 1; otherwise, the output is 0.

Boolean Expression:

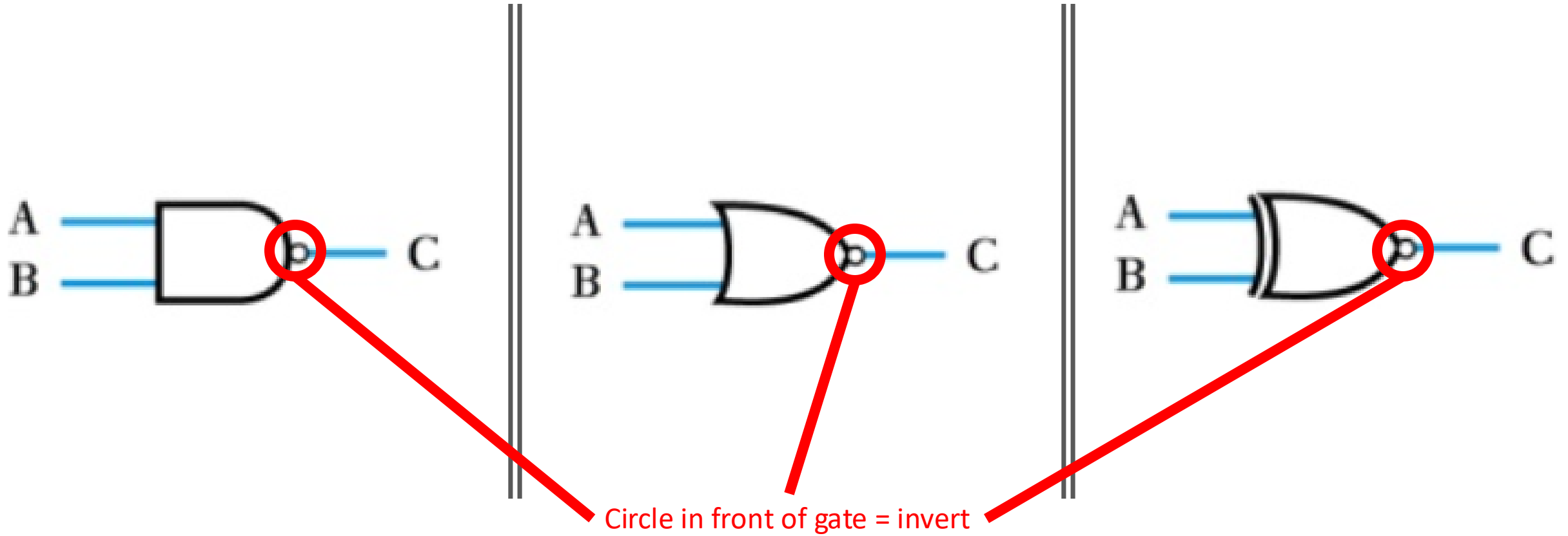
- $$X = \overline{A \oplus B} = (A \oplus B)' = \neg(A \oplus B)$$



(a) Circuit symbol

A	B	C
0	0	1
0	1	0
1	0	0
1	1	1

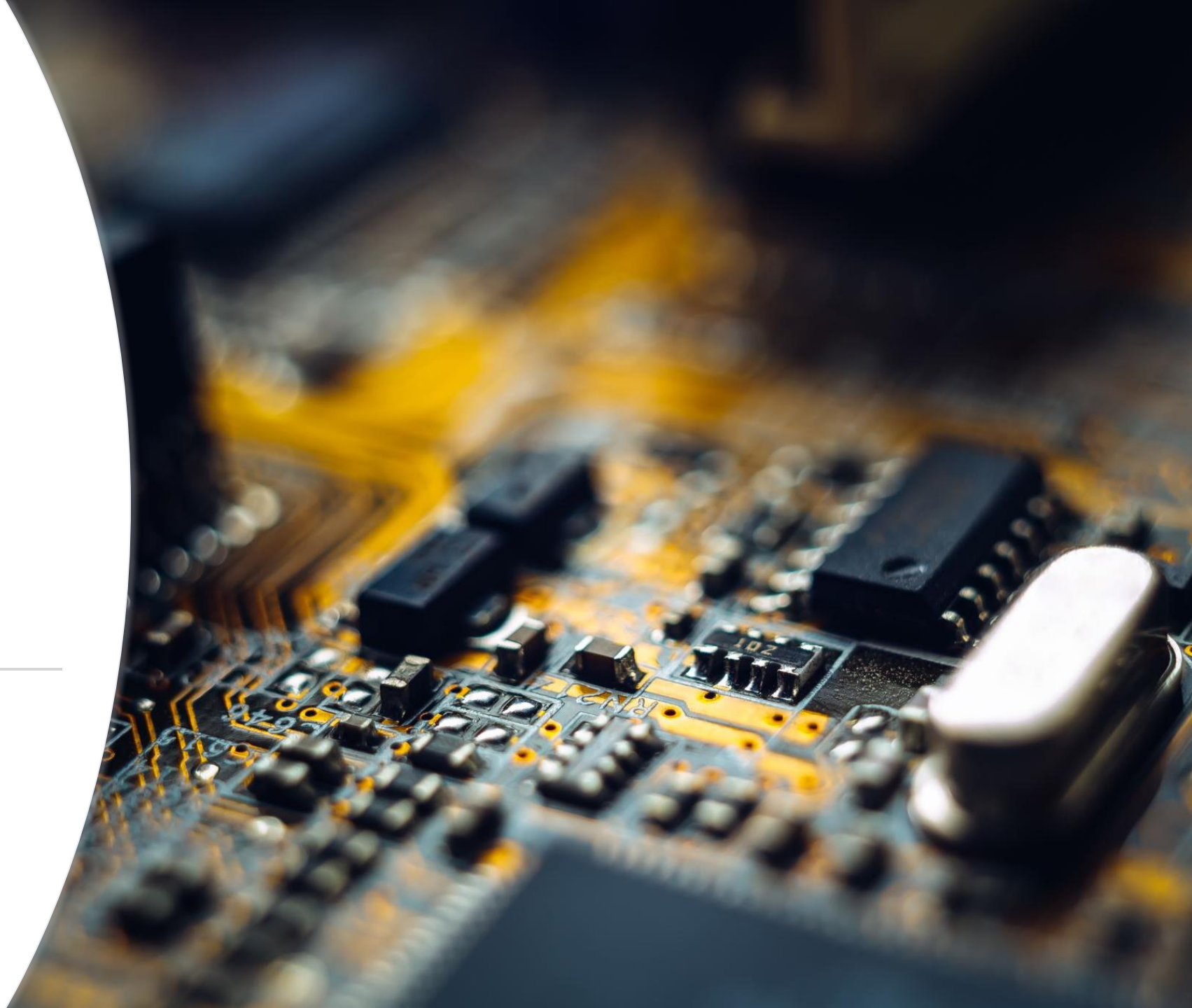
(b) Truth table



NAND – NOR – XNOR

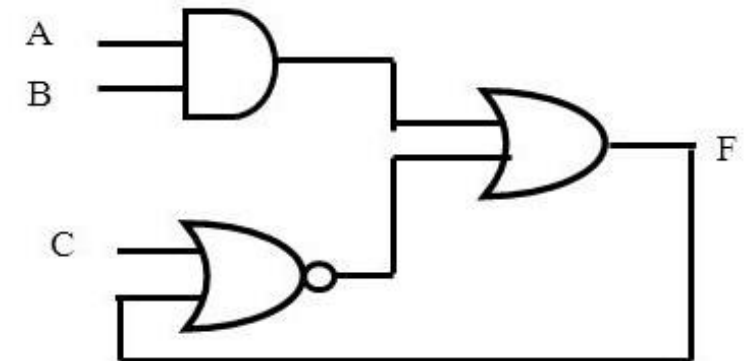
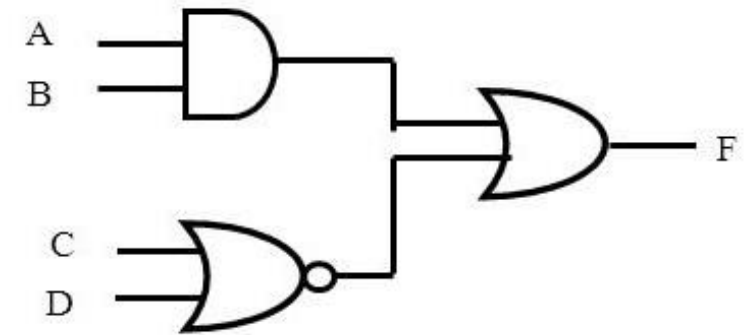


Circuits

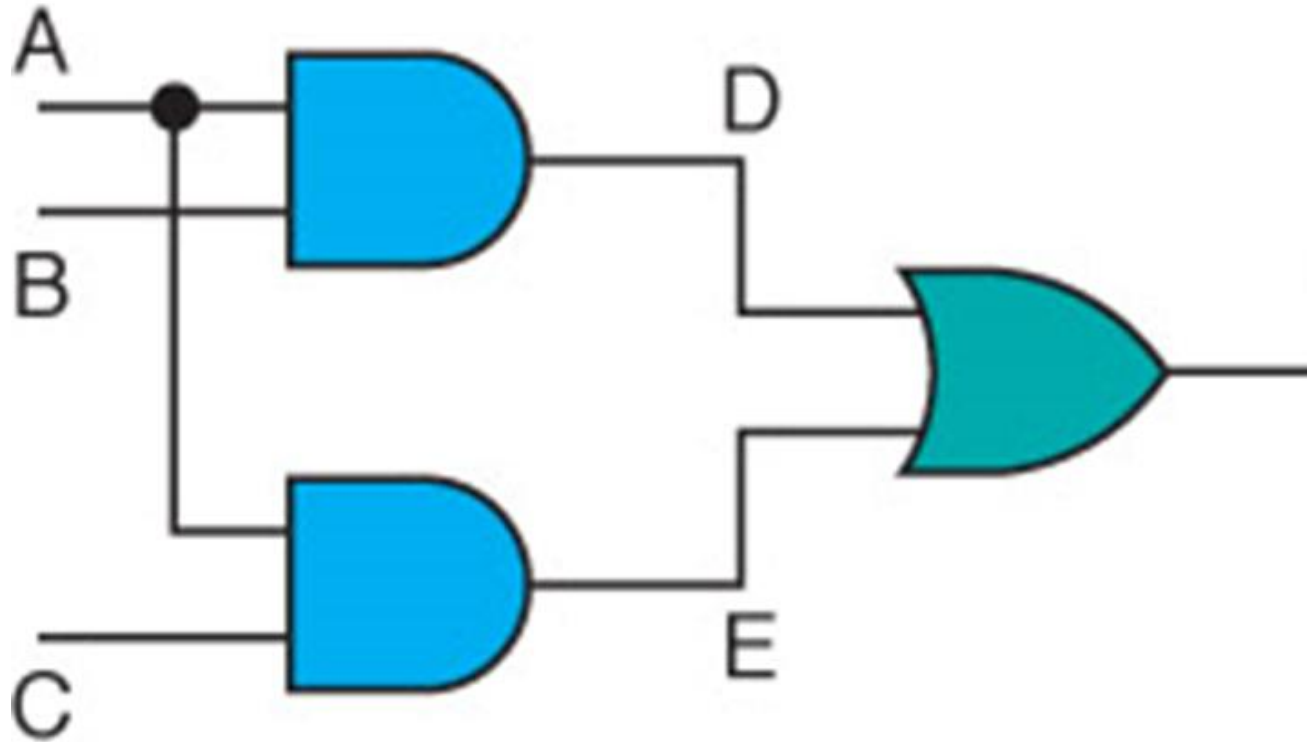


Circuits

- Combinational circuit
 - The input values explicitly determine the output
- Sequential circuit
 - The output is a function of the input values and the existing state of the circuit
- We describe the circuit operations using
 - Boolean expressions
 - Logic diagrams
 - Truth tables

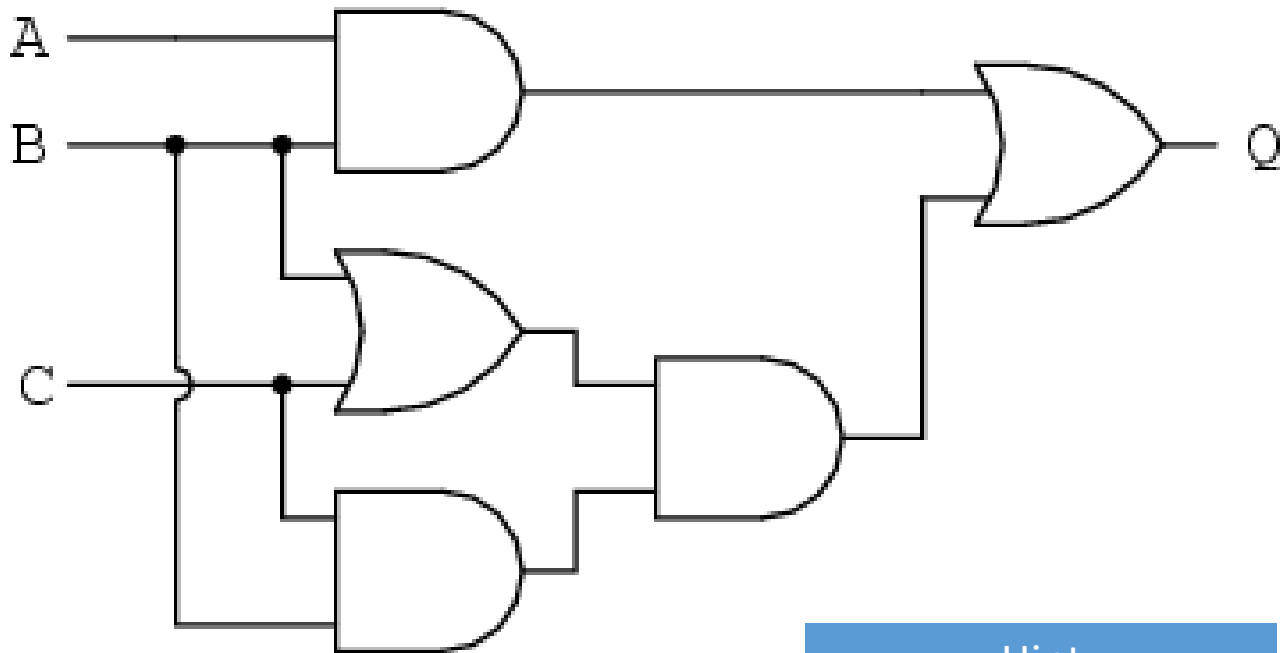


Combinational Circuits



Gates are combined into circuits by using the output of one gate as the input for another

Example 1

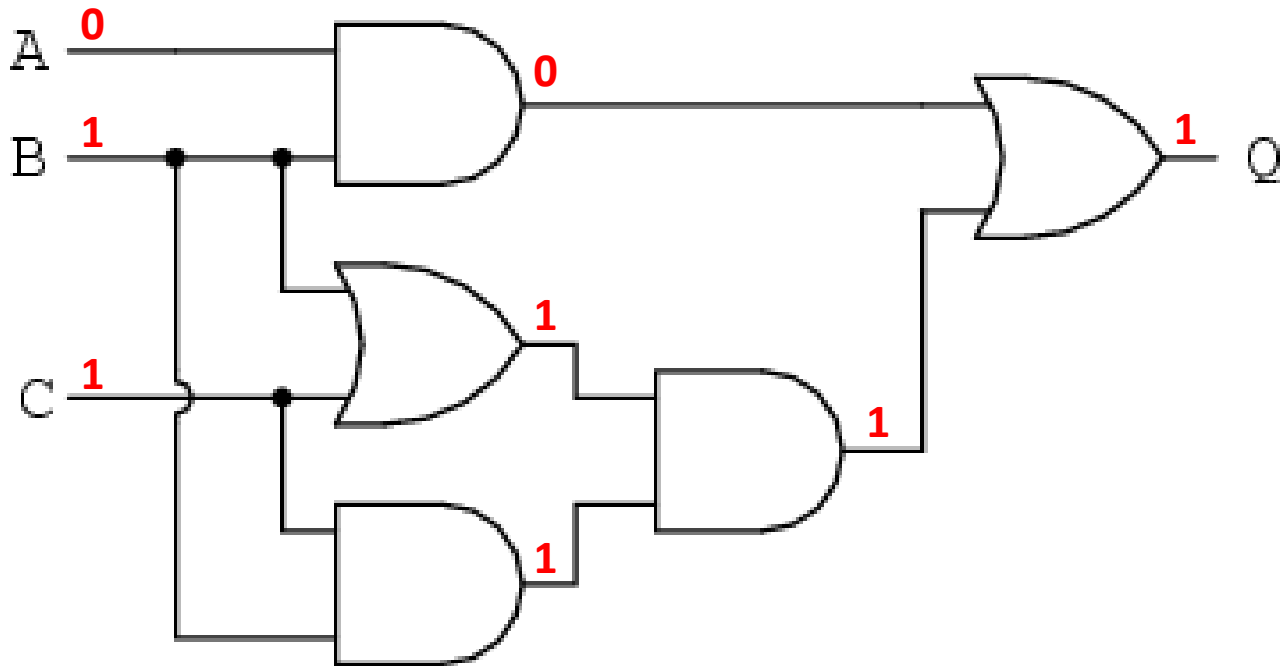


Hint
Inputs = $2^{\# \text{ variables}}$

What are all the possible inputs?

A	B	C	Q (Output)
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

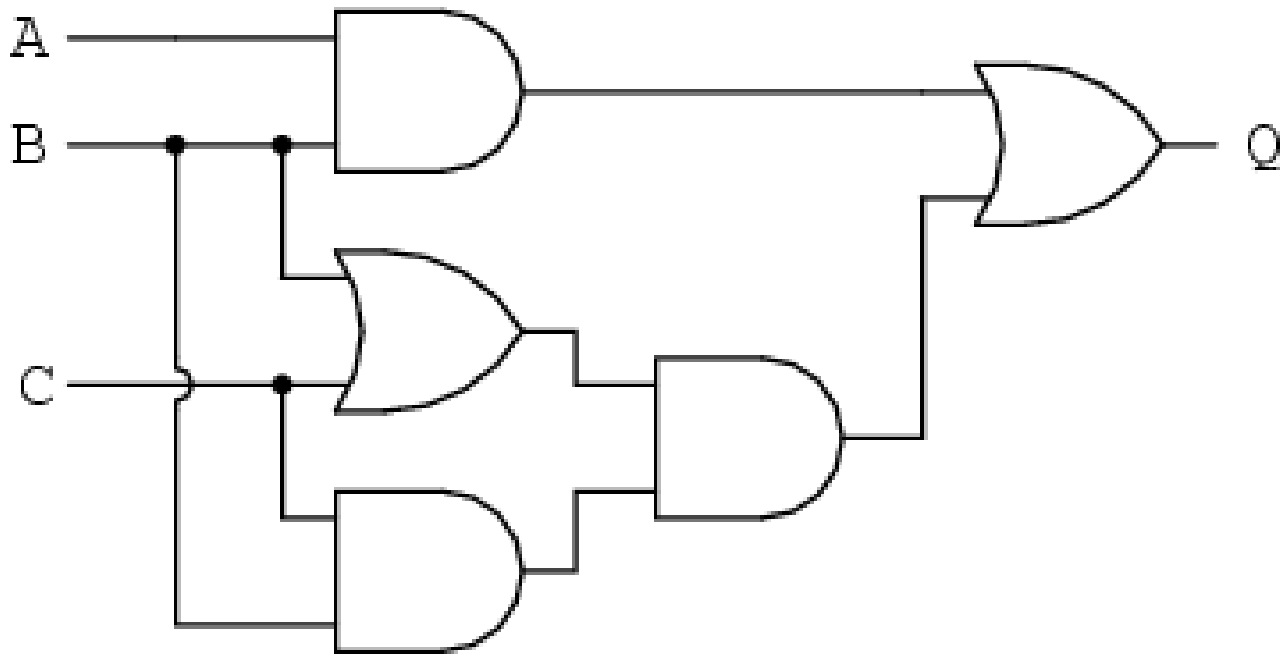
Example 1



Calculate all possible outcomes.

A	B	C	Q (Output)
0	0	0	
0	0	1	
0	1	0	
0	1	1	1
1	0	0	
1	0	1	
1	1	0	
1	1	1	

Example 1



Solution

A	B	C	Q (Output)
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

Boolean Algebra Rules

Axioms of Boolean Algebra

- Boolean algebra is based on a set of rules that are derived from a small number of basic assumptions – **axioms**
- Basic assumption: Boolean algebra, B , involves elements that take on one of two values, 0 and 1.
- The following axioms are assumed to be true...



Axioms of Boolean Algebra

$$1a. \quad 0 \cdot 0 = 0$$

$$1b. \quad 1 + 1 = 1$$

$$2a. \quad 1 \cdot 1 = 1$$

$$2b. \quad 0 + 0 = 0$$

$$3a. \quad 0 \cdot 1 = 1 \cdot 0 = 0$$

$$3b. \quad 1 + 0 = 0 + 1 = 1$$

$$4a. \quad \text{If } x = 0, \text{ then } \bar{x} = 1$$

$$4b. \quad \text{If } x = 1, \text{ then } \bar{x} = 0$$

$$5a. \quad x \cdot 0 = 0$$

$$5b. \quad x + 1 = 1$$

$$6a. \quad x \cdot 1 = x$$

$$6b. \quad x + 0 = x$$

$$7a. \quad x \cdot x = x$$

$$7b. \quad x + x = x$$

$$8a. \quad x \cdot \bar{x} = 0$$

$$8b. \quad x + \bar{x} = 1$$

$$9. \quad \bar{\bar{x}} = x$$

Single Variable Rules (Theorems)

- From the axioms, we define rules (theorems) for dealing with single variables
- We can prove by *perfect induction* (substitute $x = 0$ and $x = 1$ in each expression and use axioms)

Two- and Three- Variable Properties

$$10a. \quad x \cdot y = y \cdot x$$

Commutative

$$10b. \quad x + y = y + x$$

$$11a. \quad x \cdot (y \cdot z) = (x \cdot y) \cdot z$$

Associative

$$11b. \quad x + (y + z) = (x + y) + z$$

$$12a. \quad x \cdot (y + z) = x \cdot y + x \cdot z$$

Distributive

$$12b. \quad x + y \cdot z = (x + y) \cdot (x + z)$$

$$13a. \quad x + x \cdot y = x$$

Absorption

$$13b. \quad x \cdot (x + y) = x$$

$$14a. \quad x \cdot y + x \cdot \bar{y} = x$$

Combining

$$14b. \quad (x + y) \cdot (x + \bar{y}) = x$$

$$15a. \quad \overline{x \cdot y} = \bar{x} + \bar{y}$$

DeMorgan's theorem

$$15b. \quad \overline{x + y} = \bar{x} \cdot \bar{y}$$

$$16a. \quad x + \bar{x} \cdot y = x + y$$

$$16b. \quad x \cdot (\bar{x} + y) = x \cdot y$$

$$17a. \quad x \cdot y + y \cdot z + \bar{x} \cdot z = x \cdot y + \bar{x} \cdot z$$

Consensus

$$17b. \quad (x + y) \cdot (y + z) \cdot (\bar{x} + z) = (x + y) \cdot (\bar{x} + z)$$

Order of Operations

P

Parentheses

N

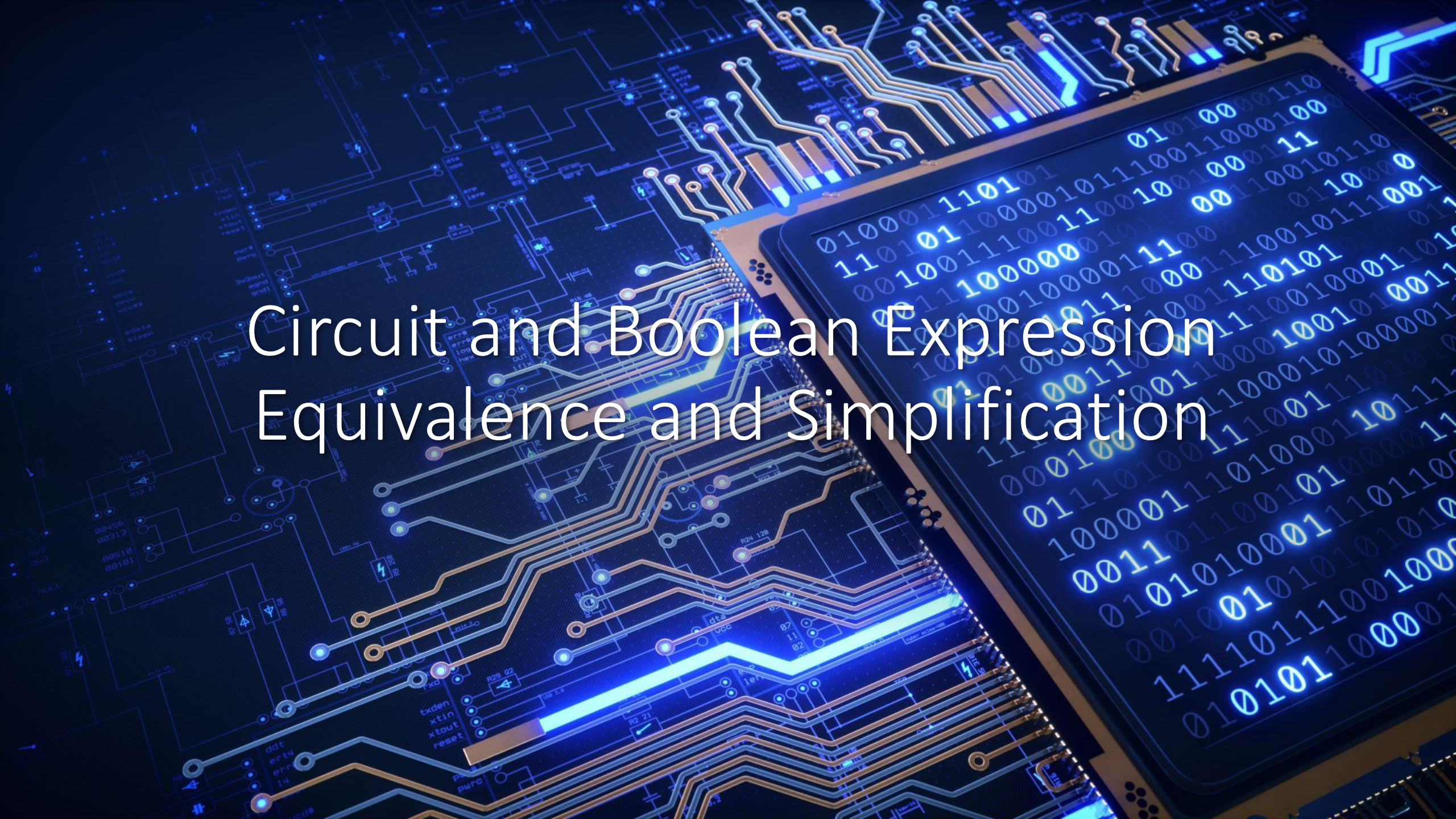
NOT operator (Complement)

A

AND operator

O

OR operator

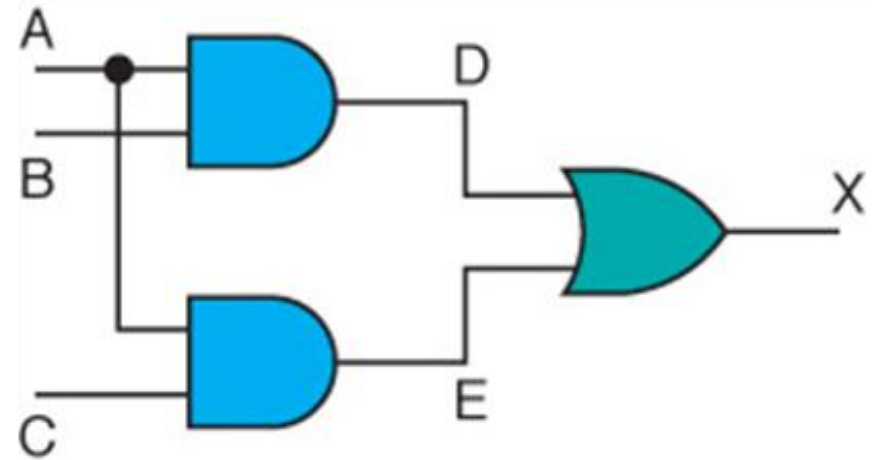


Circuit and Boolean Expression Equivalence and Simplification

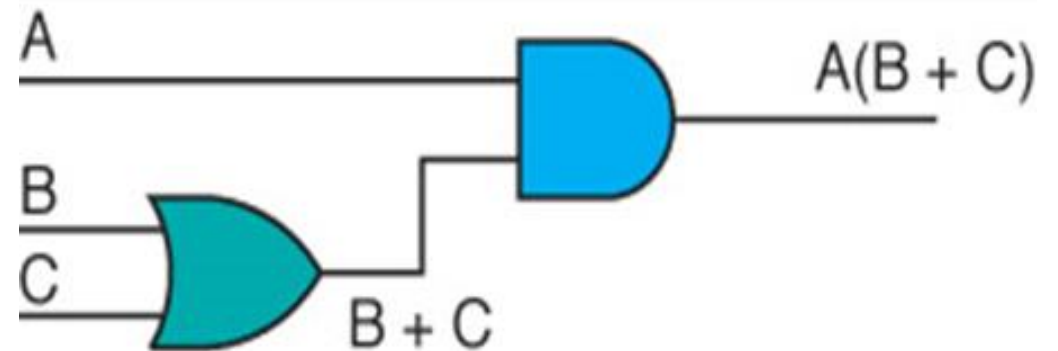
Circuit Equivalence

- Two circuits that produce the same output for identical input
- Boolean algebra allows us to apply provable mathematical principles to help design circuits
- $A(B + C) = AB + AC$ (distributive property) so circuits must be equivalent

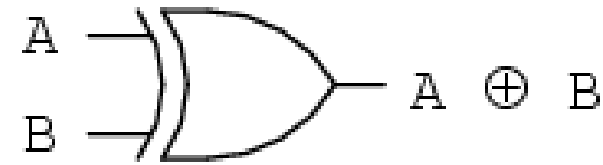
$AB + AC$



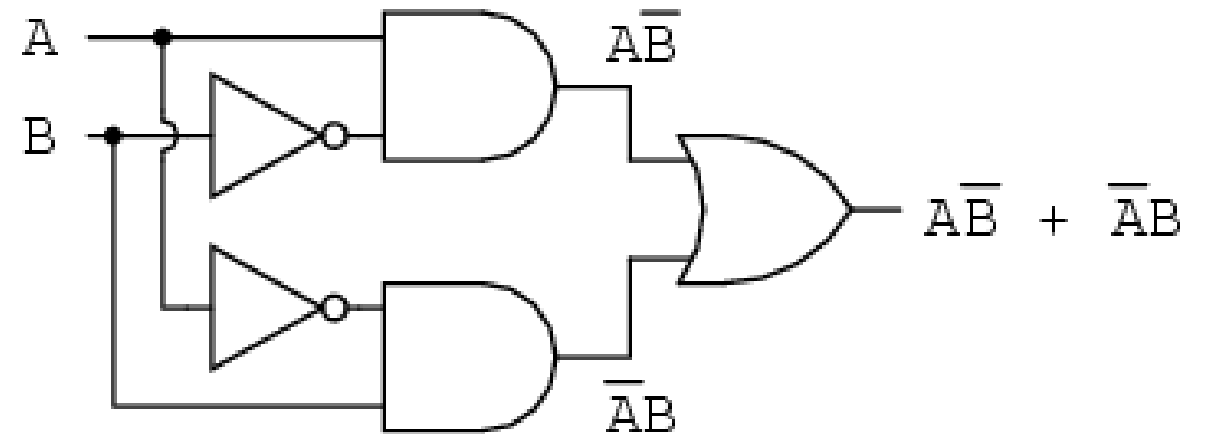
$A(B + C)$



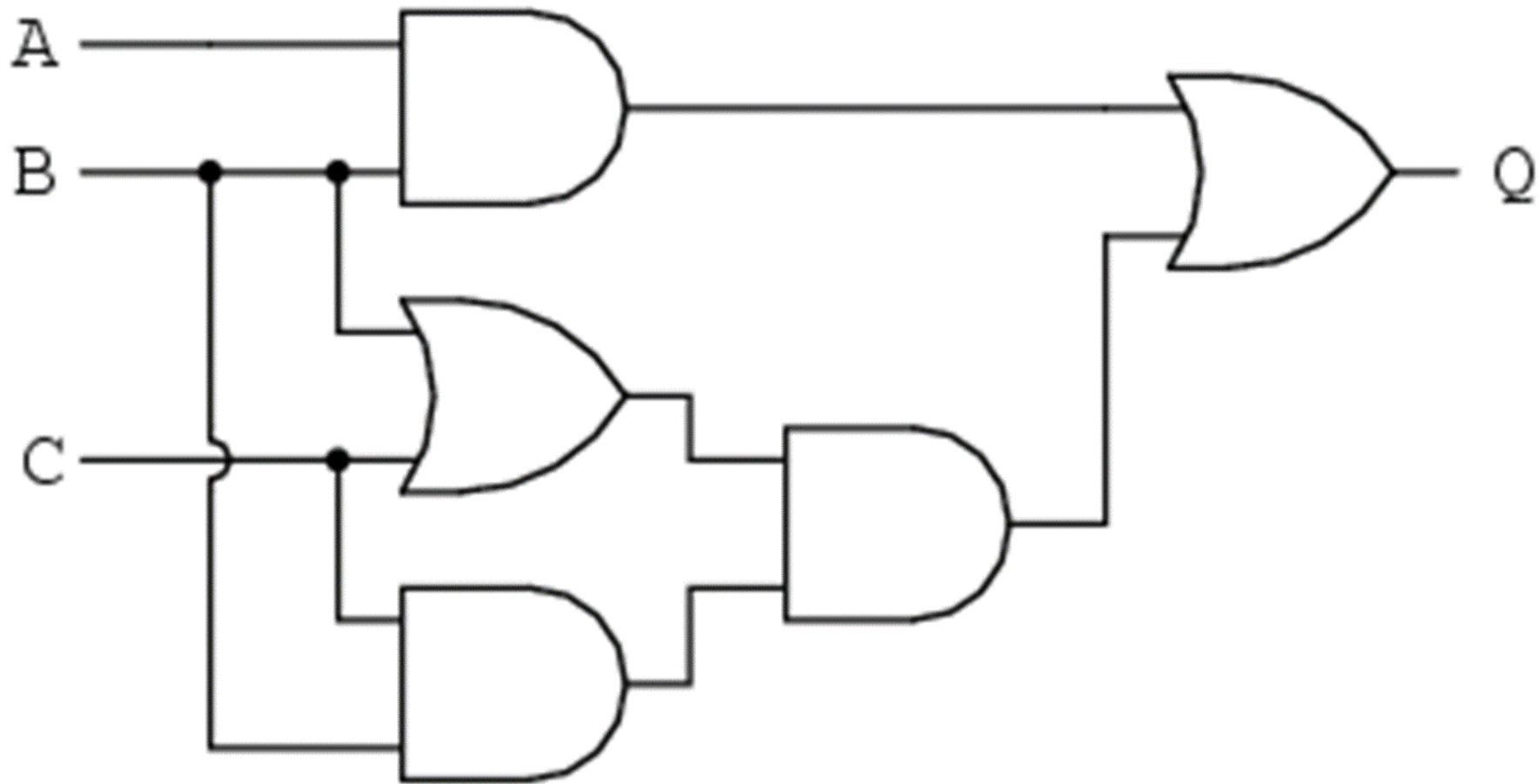
XOR Gate Equivalence



... is equivalent to ...

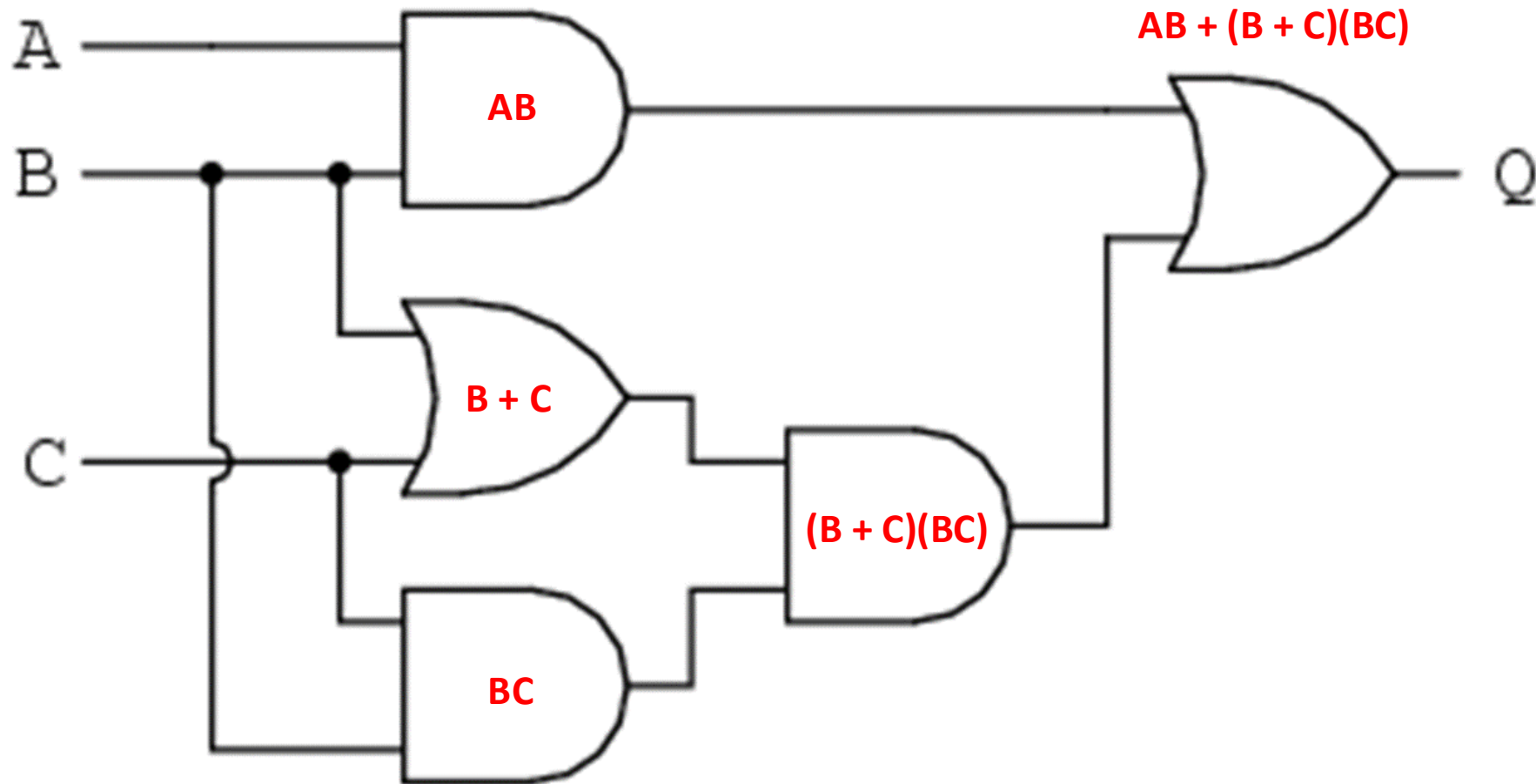


$$\mathbf{A \oplus B = A\bar{B} + \bar{A}B}$$



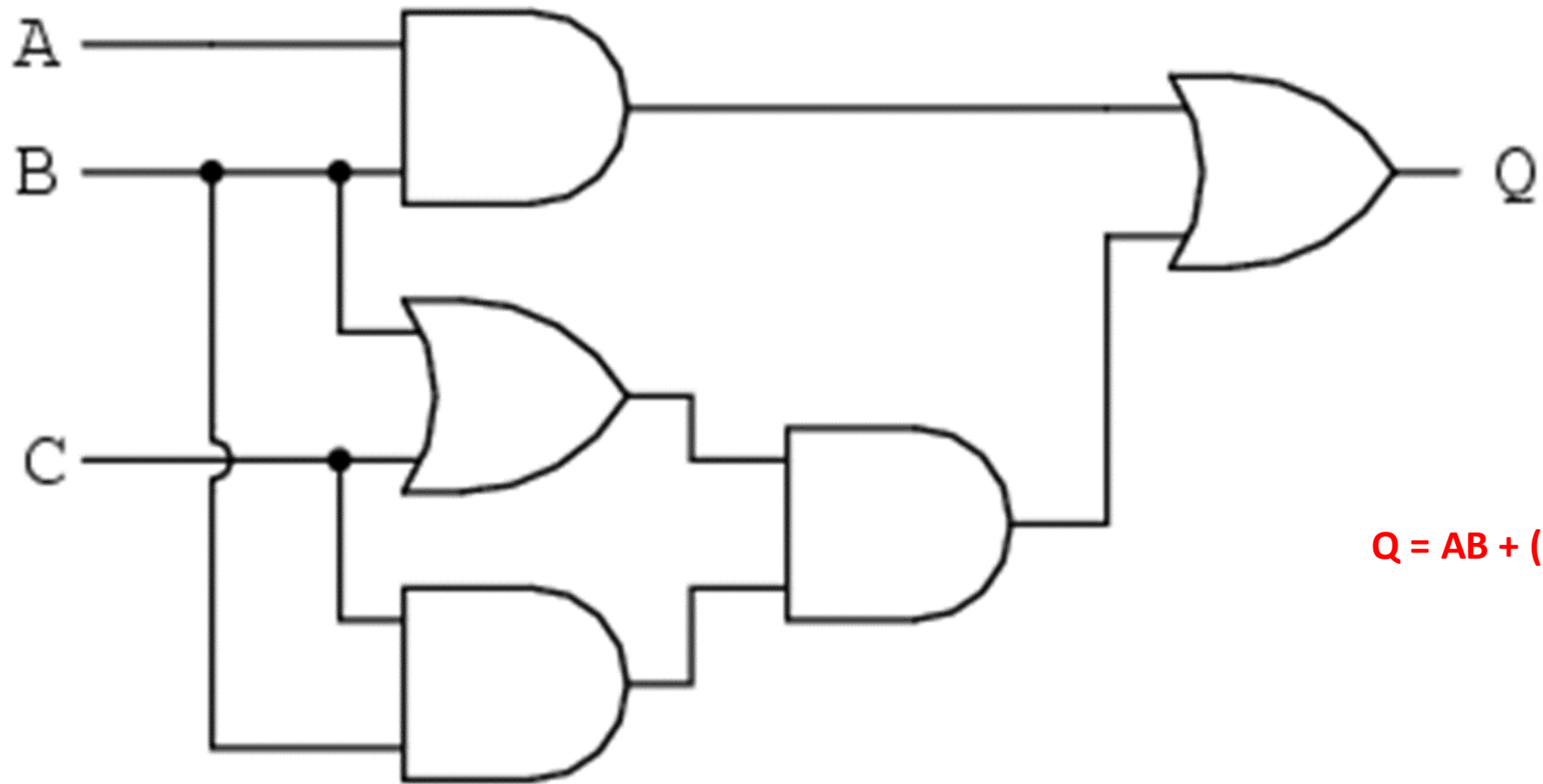
Example 2 – Part 1

Write a Boolean expression for this circuit.



Example 2 – Part 1

Start from the left to the right (from the variables to the output).

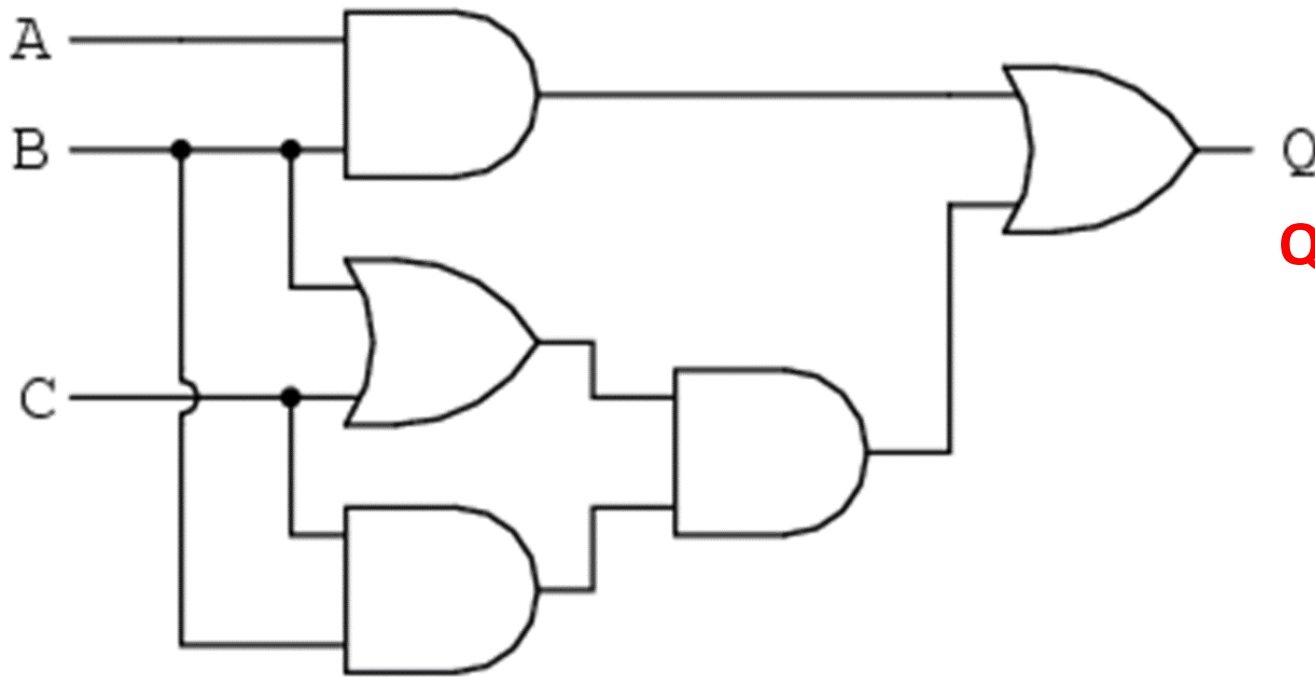


$$Q = AB + (B + C)(BC)$$

Example 2 – Part 1

Solution

Example 2 – Part 2



Simplify the Boolean expression.

$$\begin{aligned} Q &= AB + (B + C)(BC) && \text{Commutative Law} \\ &= AB + (BC)(B + C) && \text{Distribute} \\ &= AB + \cancel{BBC} + \cancel{BCC} \\ &= AB + BC + \cancel{BC} && \text{Idempotent Law (OR)} \\ &= \cancel{AB} + \cancel{BC} && \text{Factor} \\ &= B(A + C) \end{aligned}$$

Example 2 – Part 3

Create a circuit of the simplified Boolean expression.

$$Q = B(A + C)$$



A —

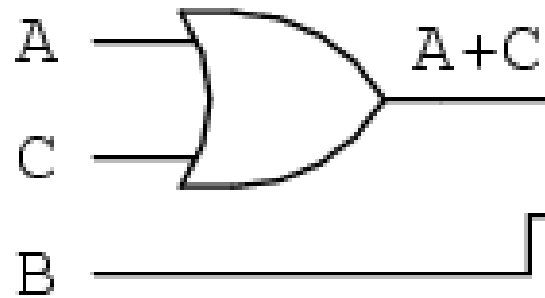
C —

B —

$$Q = B(A + C)$$

Example 2 — Part 3

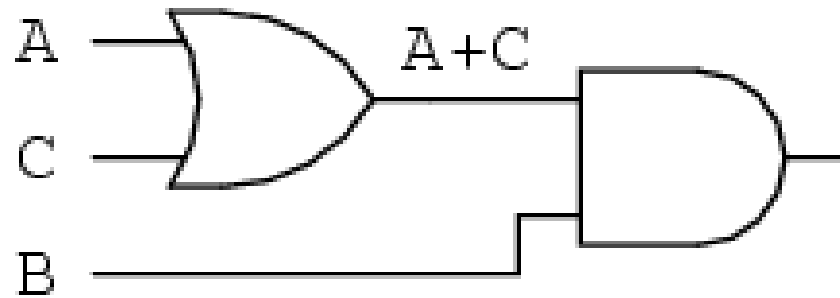
Start with the inputs



$$Q = B(A + C)$$

Example 2 – Part 3

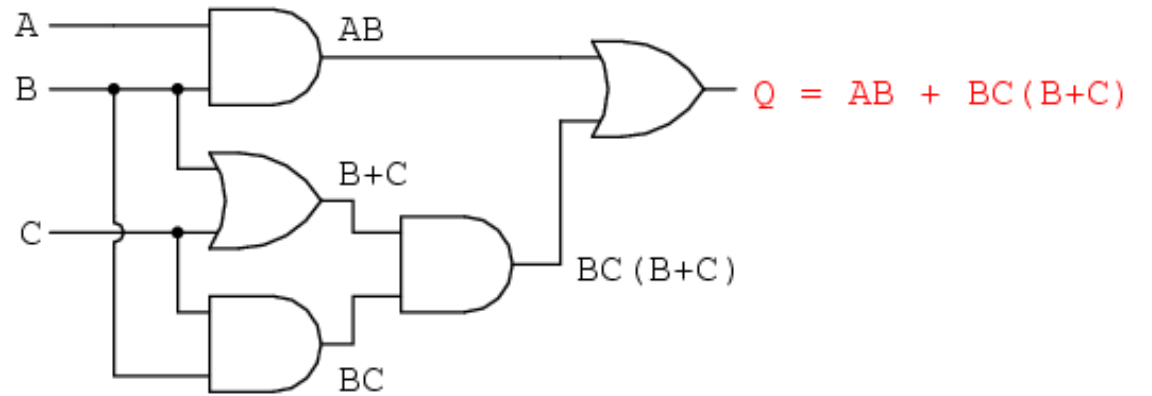
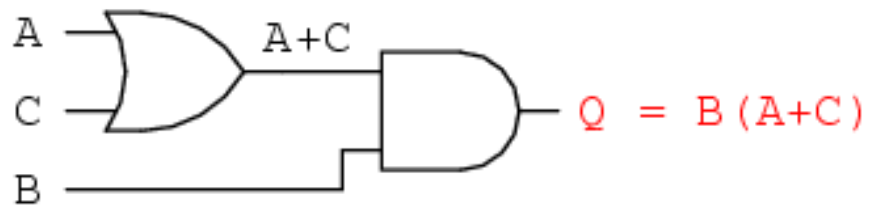
Add the first gate according to the order of operations (parentheses).



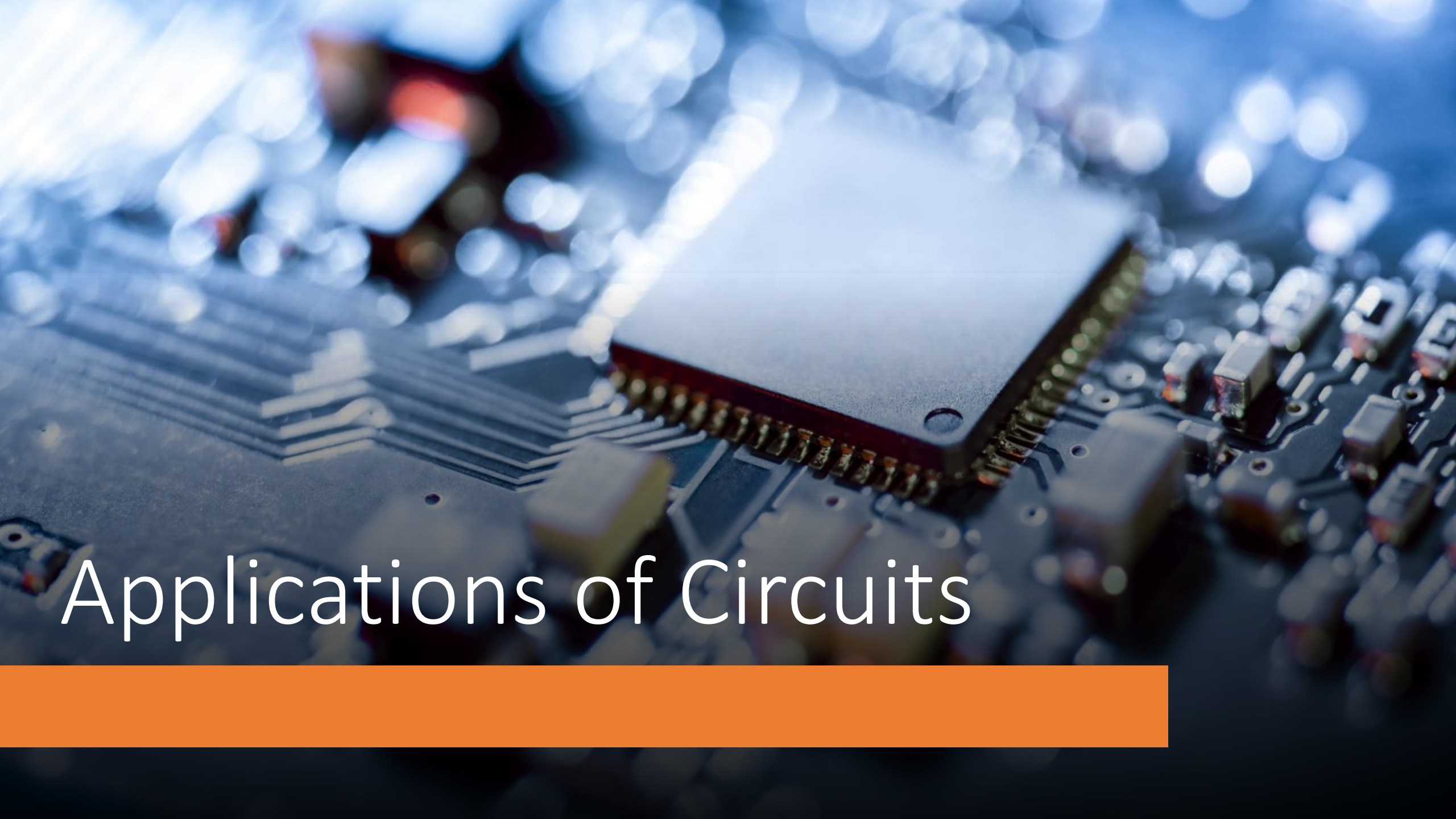
$$Q = B(A + C)$$

Example 2 – Part 3

Add the next gate.



Example 2 – Compare Circuits

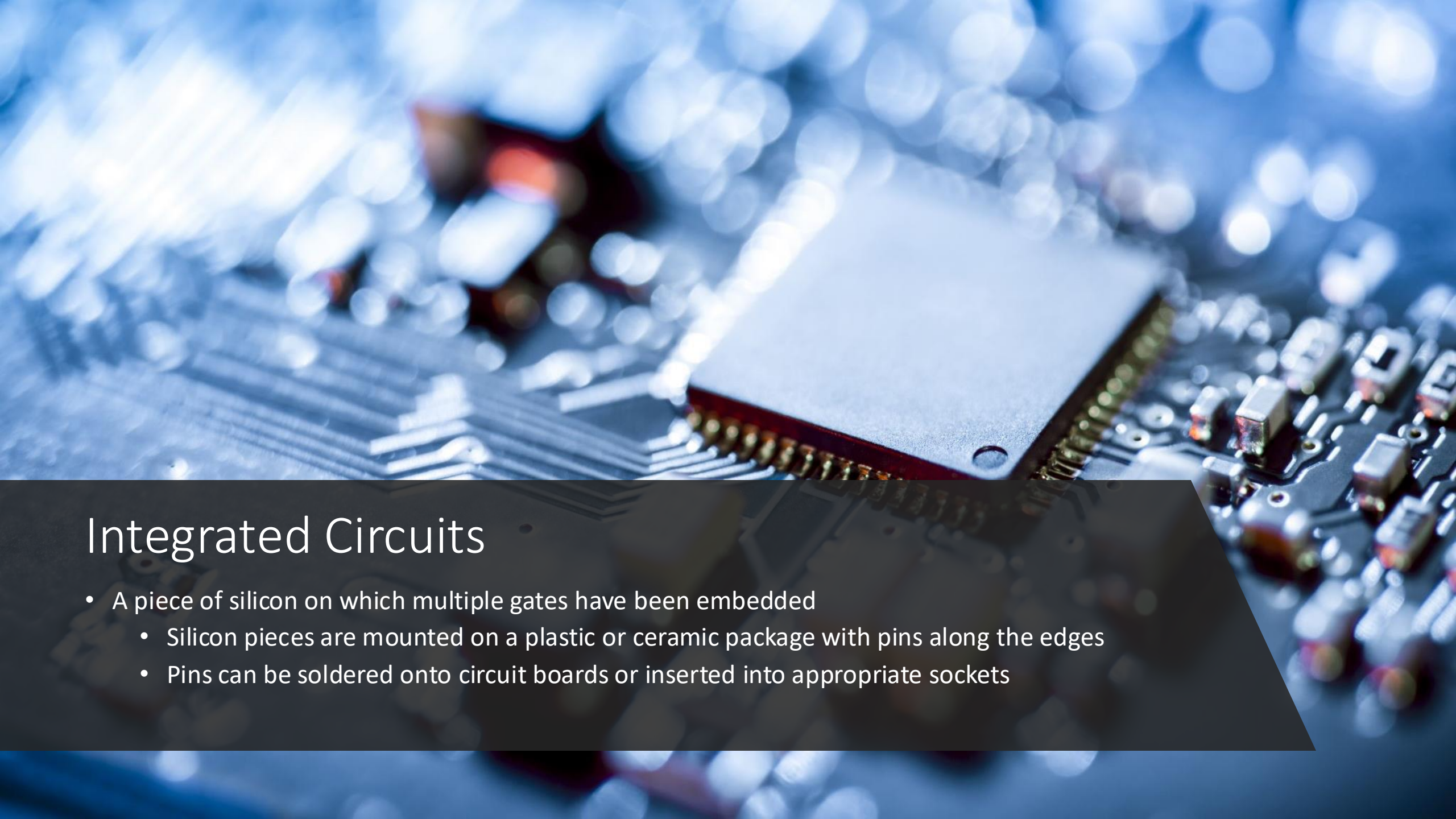


Applications of Circuits



Constructing Gates

- Gates are constructed with **transistors**
(**Transistor count**: http://en.wikipedia.org/wiki/Transistor_count)
 - A device that acts either as a wire that conducts electricity or as a resistor that blocks the flow of electricity, depending on the voltage level of an input signal
- The easiest gates to create are the **NOT**, **NAND**, and **NOR** gates
- You will learn more about this in your digital systems course



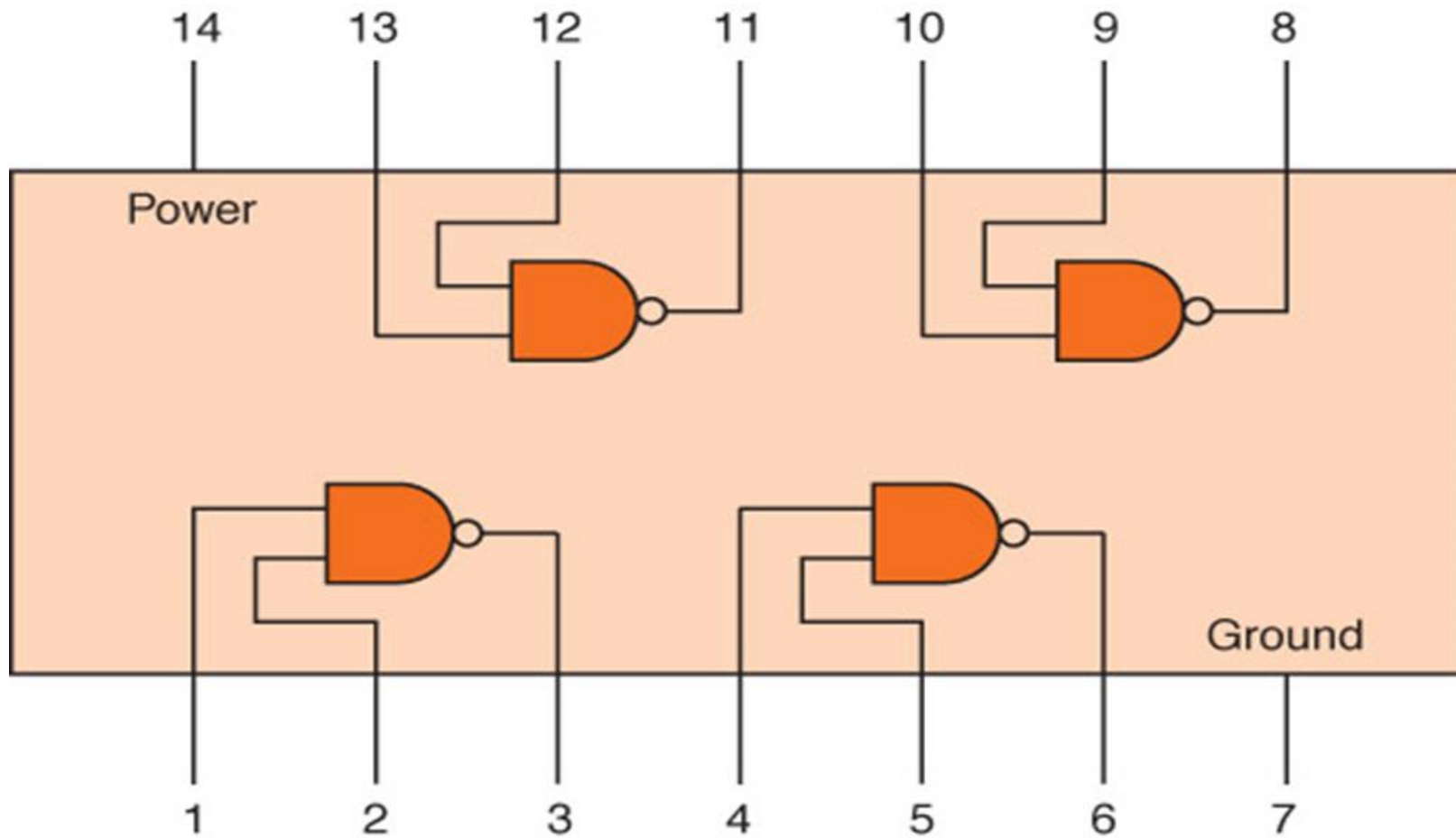
Integrated Circuits

- A piece of silicon on which multiple gates have been embedded
 - Silicon pieces are mounted on a plastic or ceramic package with pins along the edges
 - Pins can be soldered onto circuit boards or inserted into appropriate sockets

Abbreviation	Name	Number of Gates
SSI	Small-Scale Integration	1 to 10
MSI	Medium-Scale Integration	10 to 100
LSI	Large-Scale Integration	100 to 100,000
VLSI	Very-Large-Scale Integration	more than 100,000

Integrated Circuits

Integrated circuits (IC) are classified by the number of gates contained in them



Integrated Circuits

Example SSI chip
containing independent
NAND gates

CPU Chips

- The most important integrated circuit in any computer is the **Central Processing Unit**, or CPU
- Each CPU chip has many pins through which essentially all communication in a computer system occurs

